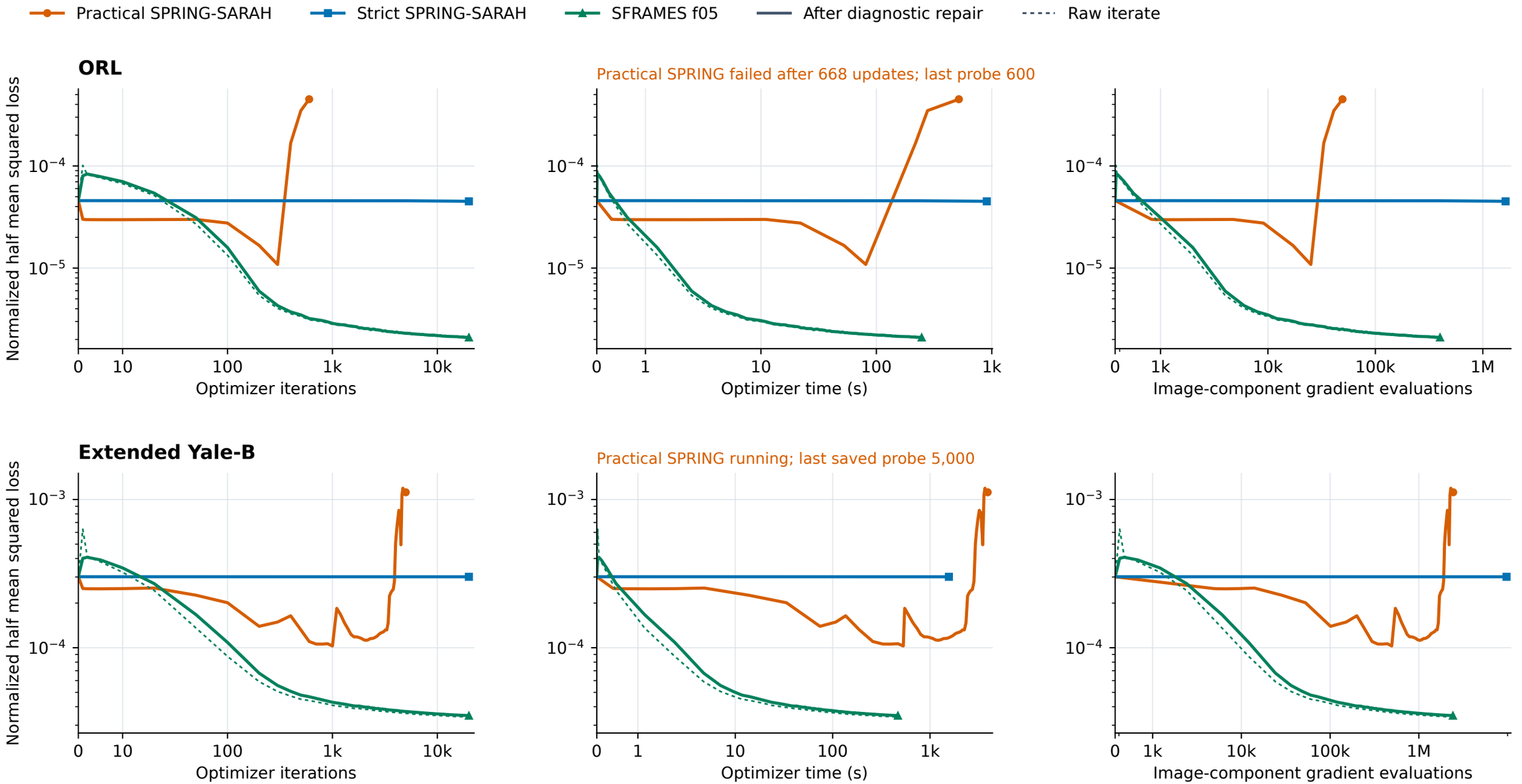
Loss by iteration, time and gradient budget

Same-seed development comparison • seed 110 • Elite snapshot at 15:20 CET, 04 November 2026

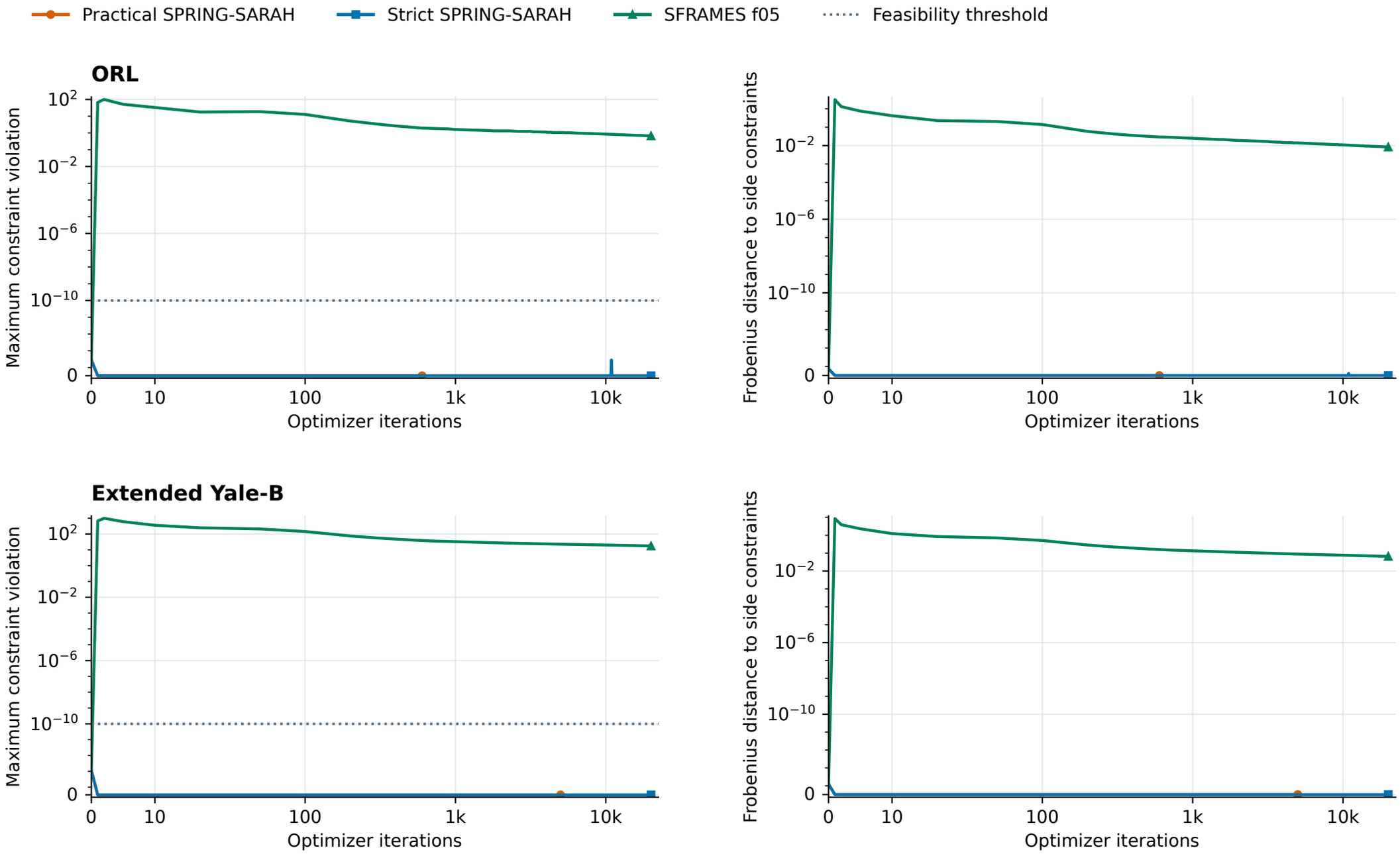


Curves stop at the last saved probe. Budget axes are linear near zero and logarithmic thereafter. Raw and repaired curves can coincide.

Optimizer time includes curvature estimation and projections. Shared-host load varied between runs. Diagnostic repair and probes are excluded.

Feasibility of the raw iterates

Same seed and saved horizons as the loss comparison. Projection certification is tracked separately from feasibility of a saved iterate.



Maximum combines the two spectral norm excesses, the Frobenius norm of the negative part of U, and V's entrywise $\ell 1$ excess.

The right panels show distance to the positivity and $\ell 1$ constraints. Values near zero are shown on a linear scale.